

Rapid Online Analysis of Photopolymerization Kinetics and Molecular Weight Using Diffusion NMR



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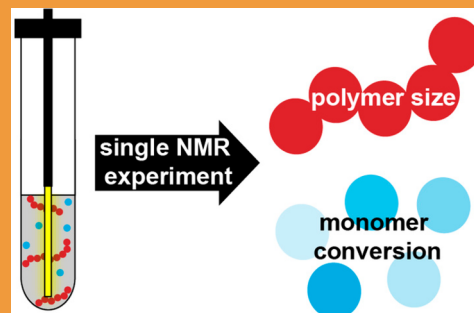


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ABSTRACT: Online, high-throughput molecular weight analysis of polymerizations is rare, with most studies relying on tedious sampling techniques and batchwise postanalysis. The ability to track both monomer conversion and molecular weight evolution in real time could underpin precision polymer development and facilitate study of rapid polymerization reactions. Here, we use a single time-resolved diffusion nuclear magnetic resonance (NMR) experiment to simultaneously study the kinetics and molecular weight evolution during a photopolymerization, with in situ irradiation inside the NMR instrument. As a model system, we used a photoinduced electron transfer reversible addition–fragmentation chain transfer (PET-RAFT) polymerization. The data allow diffusion coefficients and intensities to be calculated every 14 s from which the polymer size and monomer conversion can be extracted. Key to this approach is (1) the use of shuffled gradient amplitudes in the diffusion NMR experiment to access reactions of any rate, (2) the addition of a relaxation agent to increase achievable time resolution and, (3) a sliding correction that accounts for viscosity changes during polymerization. Diffusion NMR offers a uniquely simple, translatable handle for online monitoring of polymerization reactions.



Monitoring reactions in real time is vital for understanding reaction mechanisms and optimizing synthetic procedures. The achievable time-resolution and use of enclosed vessels enables the online analysis of rapid and highly sensitive processes, where batchwise processing is impractical. For polymerizations, online analysis of kinetics and molecular weight evolution generally requires flow reactors coupled with characterization techniques, such as gel permeation chromatography (GPC). The use of GPC limits studies to materials that fully dissolve in common solvents and confines the analysis to batchwise processing, restricting temporal resolution. Nuclear magnetic resonance (NMR) spectrometers are commonplace and are routinely used for kinetic and mechanistic studies. Inline benchtop NMR has even been used for rapid, automated screening of polymerizations. Sometimes the ^1H NMR resonances of polymer chain and end group protons can be used to estimate the apparent number-average molecular weight (M_n). However, such analysis is fraught with difficulty when used in isolation and is not possible for large polymers or where the relevant NMR signals overlap. In such instances, diffusion NMR offers an alternative means of establishing polymer size and dispersity ($\bar{D}$).

At infinite dilution, the self-diffusion coefficient (D) of any species is intrinsically linked to molecular size, shape, and solvent viscosity (η_0). For the simple case of a spherical particle with effective hydrodynamic radius r , the relationship is given by the Stokes–Einstein equation (“stick” boundary condition):

$$D = \frac{k_B T}{6\pi\eta_0 r} \quad (1)$$

where k_B is the Boltzmann constant and T is the temperature. Using the Mark–Houwink relationship for monodisperse, infinitely dilute systems, the equation can be expressed in terms of the viscosity-average molecular weight (M_v):

$$D = \frac{k_B T}{6\pi\eta_0} \left(\frac{10\pi N_A}{3KM_v^{\alpha+1}} \right)^{1/3} \quad (2)$$

where N_A is Avogadro’s number, and K and α are the Mark–Houwink coefficients specific to the polymer–solvent system. Simplify, an empirical scaling relationship is obtained:

$$D = kM_v^a \quad (3)$$

where k and a are variables specific to the polymer system that encompass the shape and density of the polymers in a given solvent and changes to solution viscosity (η) caused by polymer size. This relationship has been extended to the

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weight-average molecular weight (M_w), with calculated molecular weights compared well with those from GPC for a range of polymer samples.

Diffusion NMR measurements are based on the pulsed gradient spin-echo (PGSE) NMR sequence and variation such as the stimulated echo-based PGSTE sequence, and can be used to obtain the diffusion coefficients of polymers. Sometimes the diffusion NMR data can be presented in a 2D display mode commonly referred to as diffusion-ordered spectroscopy (DOSY) to accentuate the ability of diffusion NMR to separate components in mixtures according to their diffusion coefficients. Diffusion NMR has been used to monitor the molecular weight evolution during polymerizations via batchwise sampling of the reaction mixture. Online monitoring has been possible using a flow reactor or in a single NMR tube, though the methods were limited to polymerizations with rates that are slow compared to the diffusion NMR measurement time scale. In contrast, time-resolved diffusion NMR expands the range of reaction rates that can be studied by ensuring changes in concentration and relaxation do not impact the PGSE NMR measurement. The method has been successfully applied to a variety of dynamic systems, including oligomer formation, polymer growth, and polysaccharide degradation.

Photoinduced electron transfer–reversible addition–fragmentation chain transfer (PET-RAFT) polymerization is a controlled radical process, now routinely used to prepare vinyl polymers with diverse architectures, such as the reaction shown in Figure 1a. The method is widely used due to excellent tolerance to oxygen, spatiotemporal control, and chain-end fidelity. Despite examples of online monitoring of kinetics using Fourier transform near-infrared or NMR, no method exists for rapid online molecular weight analysis.

Herein we describe how time-resolved diffusion NMR experiments can be used in situ to monitor the kinetics and molecular size evolution during polymerizations. We demonstrate this approach using the PET-RAFT polymerization of methyl acrylate (MA) activated by zinc tetraphenylporphyrin (ZnTPP) photocatalyst as an example system (Figure 1a,b). Sample irradiation is achieved in situ using a fiber optic cable, affording an online analysis technique that is broadly applicable to any polymerization with a low dispersity.

We aimed to simultaneously measure the monomer conversion and polymer molecular weight during a polymerization reaction using a single diffusion NMR experiment. In a PGSE sequence, the first magnetic gradient pulse (of duration δ and magnitude g) spatially encodes the transverse magnetization of each species into a helix. Diffusion in the direction of the gradient attenuates the helix. After a delay Δ , an identical but effectively negative gradient pulse is then applied to unwind the helix. Thus, Δ defines the time scale of the diffusion measurement. The acquired signal intensity (I) of each species will be attenuated by both spin relaxation over the duration of the sequence and by diffusion during Δ . Denoting the signal that would be acquired in the absence of diffusion (or equivalently, absence of a gradient pulse) as I_0 , the observed NMR signal intensity for a PGSE sequence is given by the Stejskal–Tanner equation (for a single freely diffusing species):

$$I = I_0 \exp \left[-D(\gamma \delta g)^2 \left(\Delta - \frac{\delta}{3} \right) \right] = I_0 \exp[-Db] \quad (4)$$

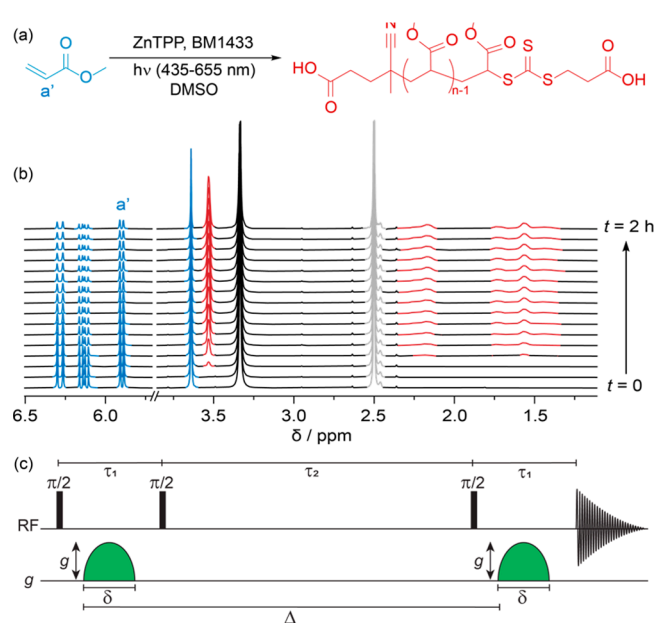


Figure 1. PET-RAFT polymerization of methyl acrylate (MA) to poly(methyl acrylate) (PMA) using photocatalyst zinc tetraphenylporphyrin (ZnTPP) and RAFT agent 4-(((2-carboxyethyl)thio)carbonothioylthio)-4-cyanopentanoic acid (BM1433) in DMSO- d_6 . (a) Reaction scheme and (b) time-dependence of diffusion NMR spectra (^1H , 500 MHz, PGSTE, $\delta = 6$ ms, $g = 2.65$ G cm^{-1} , $\Delta = 50$ ms). Reaction conditions: $[\text{MA}]/[\text{BM1433}]/[\text{ZnTPP}]$: 100:1:0.01, 0.5 M MA, 170 μM Gd-DTPA, irradiation with 567 nm, 298 K. (c) The PGSTE pulse sequence used for diffusion NMR experiments, radiofrequency (RF) pulses and gradient pulses (g), defining time periods τ_1 and τ_2 , gradient amplitude g , gradient duration δ , and diffusion time Δ . The relaxation delay (not shown) is the time between the end of one application of the pulse sequence to the start of the next. See Figure S1 for full pulse sequence used in this work.

where γ is the gyromagnetic ratio. As shown, all the gradient parameters can be summarized by the diffusion weighting factor b . In the absence of spin relaxation changes, I_0 is proportional to the concentration of the species.

Normally a series of spectra are collected with b varied monotonically by altering g to achieve a signal attenuation of $\sim 90\%$ with the largest b value used (i.e., $bD \sim 2.3$). Estimates for D and I_0 can then be obtained by nonlinear regression of $\ln(I/I_0)$ onto the data (Figure 1b, right). However, for dynamic systems, g should be varied in a shuffled order to reduce any correlation between gradient ordering and nondiffusion-based intensity changes, that is, changes in concentrations or relaxation rates. The use of a long shuffled list (e.g., 500 amplitudes) allows a moving fit (“windowing”) of $\ln(I/I_0)$ to be applied to a contiguously acquired set of spectra (Figure 1b). This leads to improved resolution that is crucial for studying rapid processes. In this work, a long shuffled list of 512 gradient amplitudes was created (Figure S1), and diffusion data was processed in windows of 32 b values.

Parameter optimization for diffusion NMR experiments on static polymer samples has been discussed previously. However, additional considerations are required for in situ measurements. For instance, it is useful to study both the monomer and the polymer during a polymerization, but the diffusion coefficients of these species may differ by orders of

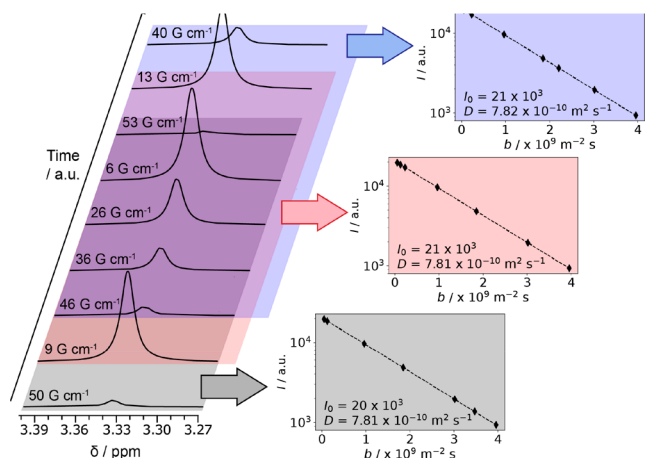


Figure 2. Principles of time-resolved diffusion NMR. Nine gradient slices from a diffusion NMR experiment acquired with shuffled g values during the PET-RAFT polymerization are displayed (NB only the water peak is shown). A moving fit to temporally contiguous subsets (three are shown) allows improved time-resolution of the estimated D and I_0 . Shuffling g values removes the correlation between the gradient order and nondiffusion-based intensity changes in dynamic systems.

magnitude due to the vastly different molecular sizes. Thus, $\sim 90\%$ attenuation will be unattainable for both species over the same range of gradient amplitudes. To tackle this issue, δ and Δ were chosen such that approximately half of the gradient amplitudes in a processing window (e.g., 4 of 7 gradient amplitudes in) resulted in a measurable signal for the fast-diffusing species, while the remaining gradient amplitudes resulted in full attenuation and were excluded from processing. The shuffled gradient amplitude list was designed to ensure the diffusion coefficients of the fast-diffusing species could be accurately determined at all times. The signal for the largest polymer species was attenuated by $\sim 70\%$, but additional experiments with parameters for optimized attenuation ($bD \sim 2$, for the largest b) confirmed the diffusion coefficient was accurately estimated ().

The polymerization kinetics can be simultaneously measured directly from the calculated intensities (I_0), as long as I_0 is not affected by time-dependent spin-relaxation constants. To avoid interference from such relaxation changes during a reaction, Δ and the relaxation delay must be carefully selected. In the PGSTE NMR experiment, signal is lost to transverse relaxation (time constant T_2) during τ_1 and longitudinal relaxation (time constant T_1) during the τ_2 period (). Thus, as Δ encompasses τ_2 , we require conditions of $\Delta < T_1$ for the proton with the shortest T_1 to avoid signal loss. Additionally, the relaxation delay should be $5 \times T_1$ for the proton with the longest T_1 to ensure sufficient relaxation and maximum signal of all protons, simplifying analysis. If a T_1 is long, the required experiment time may be impractical for studying fast processes. To improve time-resolution, a paramagnetic relaxation agent like gadolinium-diethylenetriamine pentaacetic acid (Gd-DTPA) can be added to reduce the T_1 of all protons in solution, allowing shorter relaxation delays, as we demonstrate here.

We now consider the PET-RAFT polymerization of MA in a. The longest T_1 of interest in the reaction solution prior to irradiation is 5.20 ± 0.01 s for the vinyl MA signal marked a' in Figure 1 (). Adding $170 \mu\text{M}$ Gd-DTPA

reduced this T_1 to 1.94 ± 0.05 s, with no change to the diffusion coefficient (). As such, the relaxation delay was set to 10 s ($5 \times T_1$), allowing one gradient slice to be collected every 14 s (~ 7 min to collect 32 gradient slices for the first data point). Similarly, Δ was set to 50 ms after measuring the shortest T_1 of the polymer (formed after 2 h) as 600 ± 10 ms (i.e., $T_1 \gg 50$ ms;). Time-resolved T_1 measurements () during the PET-RAFT reaction confirmed these parameters would be suitable throughout the polymerization ().

With the diffusion NMR experiment optimized for the system, the polymerization was carried out in a NMR tube, using a fiber optic for continuous irradiation in the spectrometer. To disrupt the convective flow known to result from heating during in situ irradiation, we used physical obstruction by packing the NMR tube with glass capillaries (see) to ensure accuracy of the diffusion coefficients. The measurement was repeated three times to assess reproducibility. The resulting diffusion coefficients and intensity data obtained during the PET-RAFT polymerization are shown in a,b. The intensity data (I_0) in b are representative of species concentration due to the careful selection of relaxation delay and diffusion time discussed above (see for full details). Thus, monomer conversion can be calculated during the reaction using the diffusion NMR data (c). The final conversion of $66 \pm 1\%$ after 2 h of irradiation was verified with the integrated signal intensity from a standard ^1H NMR experiment (dashed line,). This monomer conversion is lower than the 83% conversion from a previous report, which used similar experimental conditions. The reduced value may result from the use of capillaries to restrict transport rather than mixing the sample.

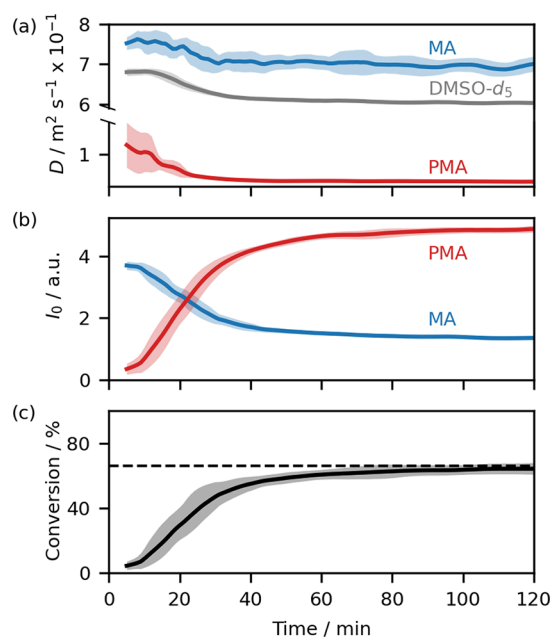


Figure 3. Simultaneous (a) diffusion, (b) intensity (I_0), and (c) conversion results from single diffusion NMR experiment (^1H , 500 MHz, PGSTE $\delta = 6$ ms, $\Delta = 50$ ms, $g = 3\text{--}50$ G cm $^{-1}$, 298 K) during PET-RAFT polymerization with in situ irradiation. Dashed line in (c) corresponds to conversion measured with ^1H NMR spectrum obtained after 2 h of irradiation. Shaded areas indicate the standard deviation over three experiments.

The diffusion coefficient of PMA after approximately 7 min irradiation was $1.2 \pm 0.6 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ and decreased to $3.2 \pm 0.1 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ after 2 h. The decrease is caused by both the increasing polymer size and the increase in solution viscosity. The viscosity change is also evident in the decreased diffusion coefficients of the monomer and residual solvent (DMSO- d_6). Both species experience an approximate 10% decrease in their respective diffusion coefficients over the 2 h experiment, consistent with a 10% increase in solution viscosity.

The diffusion coefficient of a polymer can be used to estimate its molecular weight using . A suitable calibration curve should be using standards with known molecular weights. We prepared a calibration curve using poly(ethylene glycol) (PEG) standards with peak molecular weight (M_p) of 1900–11850 g mol^{-1} (), verified by GPC (). While holds for infinitely

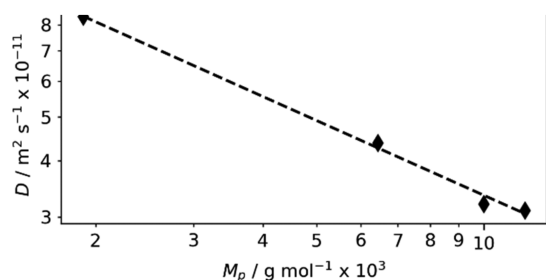


Figure 4. Calibration curve of PEG diffusing in DMSO- d_6 for estimating the molecular weight of a polymer. M_p verified with GPC and D from diffusion NMR experiments (^1H , 500 MHz, PGSTE, $\delta = 6 \text{ ms}$, $\Delta = 50 \text{ ms}$, $g = 3\text{--}50 \text{ G cm}^{-1}$, 298 K). Dashed line corresponds to nonlinear regression of onto the data.

dilute systems, the concentration of the polymer is high during the studied reaction, leading to significant interactions between molecules in solution and, thus, a higher solution viscosity. Hence, it is necessary to correct the diffusion coefficient for the observed viscosity change. From , the diffusion coefficient of each species will be equally affected by viscosity ($D \propto \eta^{-1}$), so the change in the diffusion coefficient of the solvent can be used to correct the diffusion coefficient of the polymer ().

The apparent molecular weight of the PMA polymer was estimated during the reaction for both the raw (D) and viscosity-corrected (D^*) diffusion coefficients (a), expressed relative to the equivalent weight of a PEG polymer. After 2 h of irradiation, the apparent molecular weight calculated from the diffusion coefficient was $11000 \pm 600 \text{ g mol}^{-1}$, which decreased to $8300 \pm 500 \text{ g mol}^{-1}$ using the viscosity corrected diffusion coefficient.

The standards used to create the calibration curve had a $\bar{D} < 1.05$, so the molecular weight calculated using these standards is only accurate for samples with a similar dispersity. We used GPC to measure $\bar{D}$ of 1.3 after the reaction, with M_n of $8800 \pm 200 \text{ g mol}^{-1}$ and M_w of $11500 \pm 200 \text{ g mol}^{-1}$ (). Despite the relatively high dispersity of the reaction samples, the diffusion coefficient provides a good estimate for the molecular weight and, importantly, allows the molecular weight evolution to be tracked during the polymerization.

The molecular weight-conversion plot in b shows a linear relationship, indicative of a living polymerization. The plot indicates the usefulness of the outlined diffusion NMR

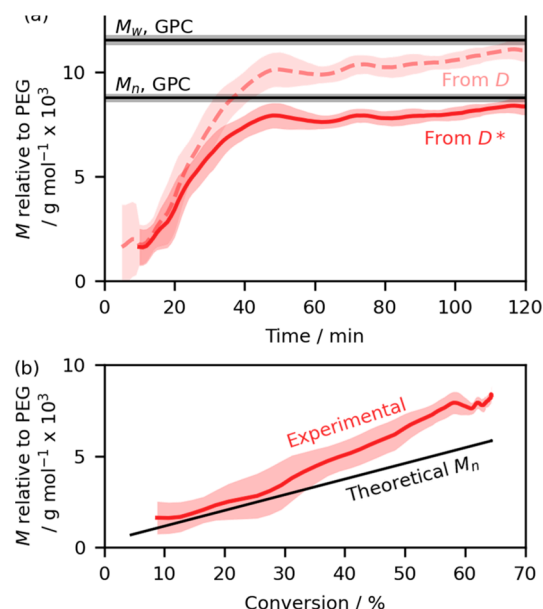


Figure 5. Molecular weight evolution during PET-RAFT polymerization. (a) Molecular weight calculated from time-resolved diffusion coefficient (raw D and viscosity-corrected D^*) using the calibration in . The horizontal lines correspond to the M_n and M_w from GPC on the same sample after 2 h of irradiation. (b) Molecular weight-conversion plot. The solid black line indicates the theoretical M_n calculated by $M_{n(\text{theoretical})} = ([\text{MA}]_0/[\text{BM1433}]_0) \times (\%C) \times (MW_{\text{MA}} + (MW_{\text{BM1433}}) \times (\%C))$, where $[\text{MA}]_0/[\text{BM1433}]_0$ = initial ratio of monomer to RAFT agent; $\%C$ = conversion = $I_{\text{polymer}}/(I_{\text{monomer}} + I_{\text{polymer}})$, I_n = intensity of ^1H NMR signal for n ; MW_{MA} and MW_{BM1433} = molecular weight of MA monomer and BM1433, respectively. Experimental parameters: ^1H , 500 MHz, PGSTE $\delta = 6 \text{ ms}$, $\Delta = 50 \text{ ms}$, $g = 3\text{--}50 \text{ G cm}^{-1}$, 298 K. Shaded areas indicate the standard deviation averaged over three experiments.

experiment, with both molecular weight and conversion data extracted from the same experiment.

To demonstrate the versatility of the proposed method, we have also studied the aqueous RAFT polymerization of acrylic acid, GPC analysis of which is challenging due to the charged systems created by the polyelectrolytes. With our method, the monomer conversion and molecular weight can also be estimated over time (see).

It is not possible to use the diffusion coefficient to measure dispersity with our method, although this has been previously demonstrated with diffusion NMR experiments. Using our diffusion NMR parameters, the NMR signal for largest polymer formed is attenuated by $\sim 70\%$ and the data are well-described by . To measure the dispersity, a deviation of the experimental data from the fit of is required to model the distribution of diffusion coefficients. Such deviations are generally only detectable at very high attenuation. Our methodology might be adapted to achieve these high levels of attenuation for the polymer, while also monitoring the faster diffusing species, by using a spectrometer capable of applying a larger maximum gradient amplitude (note in this work $g_{\text{max}} = 51 \text{ G cm}^{-1}$).

A single diffusion NMR experiment can be used to monitor polymerization kinetics and molecular weight evolution with high time-resolution (14 s between data points). This versatile method does not require any specialist equipment and can be applied to fast reaction processes in any solvent where NMR

spectra can be collected, including those that are unsuitable for GPC analysis. By eliminating the need for sequential or parallel analysis of properties, data can be efficiently collected to provide insights into highly time-sensitive polymerization reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at

Synthetic procedures, NMR experimental data, spectra, and analysis ()

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The manuscript was written through contributions of all authors.

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Notes

The authors declare no competing financial interest.

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